HR Analytics Dashboard: Project Documentation

Tool: Power BI Desktop, MySQL 8.0 | Dataset: IBM HR Analytics Employee Attrition | Prepared by: Karan Tormal | Date: 31 Oct 2026

1. Objective

Build an interactive HR analytics dashboard to understand employee attrition, identify its key drivers, and provide actionable recommendations for HR decision-making.

2. Dataset

- **Name:** IBM HR Analytics Employee Attrition
- **Size:** 1,470 rows, 35 columns
- **Key columns:** Attrition (Yes/No), Department, JobRole, OverTime, MonthlyIncome, YearsAtCompany, Gender, Age
- **Data quality:** No missing values, no duplicate Employee IDs.

3. Workflow

Step 1: Data import (MySQL Workbench)

- Created database `task4` and table `employees` with 35 columns.
- Loaded the CSV using the Table Data Import Wizard, after `LOAD DATA LOCAL INFILE` was blocked by a client-side restriction.
- Verified row count: 1,470.

Step 2: Data preparation (SQL)

- Excluded constant columns from analysis: `EmployeeCount`, `Over18`, `StandardHours`.
- Created the `hr_clean` view with an `Attrition_flag` column (Yes = 1, No = 0).
- Data quality check: 0 NULLs in key columns, 0 duplicate IDs.

Step 3: Export and load (Power BI)

- Exported `hr_clean` to CSV with "Don't Limit" set, so all 1,470 rows were exported.
- Loaded into Power BI via Get Data, then Text/CSV, and verified column types.

Step 4: KPIs (DAX measures)

```
Attrition Rate % = DIVIDE(SUM(hr_clean[Attrition_flag]), COUNTROWS(hr_clean)) * 100
Avg Monthly Income = AVERAGE(hr_clean[MonthlyIncome])
Avg Tenure (Years) = AVERAGE(hr_clean[YearsAtCompany])
```

Step 5: Filters and charts

- **Slicers:** Department, Gender, OverTime.
- **Charts:** Attrition Rate % by Department, by OverTime, and by Job Role.

4. Key Results

KPI	Value
Total employees	1,470
Employees who left	237
Attrition rate	16.12%
Avg monthly income	6,502.93
Avg tenure	7.01 years

5. Insights and Recommendations

1. **OverTime:** Employees who work overtime have an attrition rate of about 30%, compared with about 10% for those who don't.
   Recommendation: Review workload and overtime policy.
2. **Department:** Sales has the highest attrition by department (about 20.6%).
   Recommendation: Focus retention and incentive plans on the Sales team.
3. **Job Role:** Sales Representatives have the highest attrition by role (about 40%).
   Recommendation: Improve career growth paths and onboarding for this role.

6. Testing

Test	Expected	Result
Row count (SQL and Power BI)	1,470	Matched
Attrition sum (SQL and Power BI)	237	Matched
Attrition rate (overall)	16.1 / 16.12	Matched
Sales + OverTime = Yes filter	128 employees, 48 leavers	Matched in SQL and Power BI
Filters reset to overall values	1,470 / 237 / 16.12	To confirm

7. Limitations

- The dataset is a sample (IBM HR), not real company data, so insights are illustrative.
- Attrition is a binary Yes/No flag and does not capture reasons for leaving.
- Correlation shown in charts does not prove causation. For example, overtime may be linked to other factors.

8. Deliverables

- Power BI file: `task4.pbix`
- SQL scripts: table creation, import, and `hr_clean` view
- Dashboard screenshot
- This documentation